

BRETON PLOTS, AB, Canada

WMO: 712330

Lat: 53.0894N

Lon: 114.4408W

Elev: 2794

StdP: 13.27

Time Zone: -7.00 (NAM)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.8	-15.3	-30.9	1.1	-21.0	-23.4	1.7	-14.1	16.4	28.5	14.2	27.8	2.4	150	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	82.4	63.5	78.8	61.5	75.7	59.9	67.0	78.0	64.4	74.5	62.1	72.3	6.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.1	95.8	72.1	60.6	87.5	69.7	58.2	80.3	66.6	33.2	78.3	31.1	74.8	29.3	72.7	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.6	13.3	11.5	-28.8	88.2	6.1	4.5	-33.2	91.4	-36.8	94.1	-40.2	96.6	-44.6	99.8		
			-28.5	70.4	5.6	3.5	-32.5	72.9	-35.8	74.9	-38.9	76.9	-43.0	79.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.0	16.6	19.9	26.8	39.3	49.7	56.9	61.7	59.5	51.3	40.4	27.1	17.7
	DBStd	19.48	16.79	14.08	13.58	9.50	7.97	5.67	5.66	5.81	7.75	8.56	12.10	14.91
	HDD50	5126	1036	844	720	333	103	8	1	3	76	313	688	1001
	HDD65	9545	1501	1264	1185	770	474	252	134	187	414	761	1137	1466
	CDD50	1116	0	0	1	13	96	214	363	298	114	16	1	0
	CDD65	60	0	0	0	0	1	8	32	17	2	0	0	0
	CDH74	1071	0	0	0	3	59	146	445	311	96	11	0	0
	CDH80	235	0	0	0	0	5	31	111	70	17	1	0	0
Wind	WSAvg	4.7	4.0	4.2	5.0	6.0	5.8	5.0	4.6	4.1	4.2	4.7	4.1	4.1
Precipitation	PrecAvg	19.3	1.0	0.6	0.7	1.2	2.2	3.4	3.7	2.5	1.5	0.9	0.8	0.7
	PrecMax	25.7	3.0	1.2	1.4	2.2	4.3	6.9	8.2	5.4	3.4	3.0	2.4	1.9
	PrecMin	12.6	0.1	0.0	0.1	0.2	0.7	1.0	1.8	1.2	0.4	0.2	0.2	0.2
	PrecStd	3.1	0.7	0.3	0.3	0.6	0.9	1.3	1.5	0.8	0.9	0.7	0.6	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.9	52.6	57.8	72.0	79.6	84.3	88.4	86.4	82.0	72.9	59.0	50.8
		MCWB	39.8	39.0	42.8	51.1	55.8	63.7	68.2	64.7	60.3	53.9	45.3	39.7
	2%	DB	46.5	46.6	51.9	66.2	75.0	77.9	82.9	81.2	76.4	63.9	51.6	45.5
		MCWB	37.0	36.4	40.3	48.3	54.4	60.0	65.7	63.0	58.2	49.9	41.0	37.1
	5%	DB	42.2	42.0	47.7	60.7	71.0	73.8	78.8	77.2	71.7	58.5	46.4	40.9
		MCWB	34.5	33.5	37.7	45.2	52.8	58.4	63.4	61.8	56.0	46.3	38.2	33.9
	10%	DB	38.0	37.9	43.7	55.7	66.4	70.4	75.2	73.3	66.5	54.2	41.8	36.9
		MCWB	31.9	31.2	35.4	42.5	50.5	56.5	61.5	59.9	53.9	44.3	35.2	31.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.7	40.2	43.8	52.7	58.9	66.2	71.6	69.0	62.5	56.0	46.6	40.6
		MCDB	49.5	50.8	54.8	68.3	74.7	80.8	84.7	80.0	75.5	70.6	57.3	49.5
	2%	WB	37.7	36.8	40.4	49.3	55.8	62.8	68.2	66.1	59.7	51.2	41.7	37.1
		MCDB	45.5	45.9	50.3	64.7	70.7	74.0	79.5	76.9	72.8	62.0	50.0	44.4
	5%	WB	35.1	34.1	38.3	46.0	53.5	60.4	65.6	63.7	57.4	47.9	38.9	34.1
		MCDB	41.6	41.2	46.5	59.0	67.5	70.5	75.2	74.1	68.8	56.9	45.7	40.2
	10%	WB	32.4	31.6	35.8	43.2	51.4	58.3	63.0	61.3	54.9	45.2	35.7	31.1
		MCDB	37.8	37.2	43.0	54.3	64.3	67.8	72.6	70.9	64.7	52.8	40.7	35.9
Mean Daily Temperature Range	5% DB	MDBR	18.9	20.6	19.7	20.7	22.4	21.3	22.1	22.4	22.6	19.7	16.7	17.3
		MCDBR	21.4	24.6	24.5	30.0	30.3	28.1	28.1	29.4	32.7	27.3	21.7	19.0
		MCWBR	14.1	16.3	14.6	15.8	13.9	13.1	13.2	14.3	16.2	15.3	13.5	13.0
	5% WB	MCDBR	20.8	23.8	23.6	28.9	27.6	24.4	25.9	26.8	29.6	26.4	20.7	18.8
		MCWBR	14.6	16.0	14.5	15.6	13.3	12.6	13.6	14.3	15.5	15.3	13.3	13.3
Clear-Sky Solar Irradiance	taub	0.250	0.257	0.296	0.328	0.342	0.350	0.350	0.357	0.311	0.281	0.253	0.239	
	taud	2.341	2.390	2.393	2.381	2.387	2.384	2.431	2.368	2.492	2.560	2.475	2.325	
	Ebn at Noon	241	277	285	288	288	285	283	274	277	266	242	222	
	Edh at Noon	18	23	29	34	35	36	34	34	26	20	16	15	
All-Sky Solar Radiation	RadAvg	310	622	1080	1465	1748	1860	1929	1598	1114	652	332	230	
	RadStd	22	45	67	128	125	130	117	107	117	69	29	18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		-1.48	N/A	N/A	-2.81	-3.96	-3.89	N/A	+506	-126	-49
Regional (60 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page